Math 15 — Unit 3: Assignments

The assignment from every lesson in the booklet. Printed work is not tracked on the website.

Lesson 1 Assignment *Show your work.*

Part A — Vocabulary and Like Terms

1. Classify each polynomial as a monomial, binomial, or trinomial.
(a) $4x + 9$ (b) $6x^3$ (c) $x^2 - 5x + 2$

2. State the degree of each term.
(a) $7x$ (b) $3x^2$ (c) -9 (d) $5xy$ (e) x^3

3. For the polynomial $6x^3 - 4x^2 + x - 8$, state:
(a) the number of terms. (b) the coefficient of each variable term. (c) the constant term. (d) the degree of the polynomial.

4. Sort into groups of like terms: $3x$, $-5x^2$, 8, x, $-2x^2$, $4y$, -9, $6y$.
x-terms: x^2-terms: y-terms: Constants:

5. State whether each pair is a pair of like terms. If not, explain why not.
(a) $4x$ and $9x$ (b) $3x^2$ and $3x$ (c) $5xy$ and $-2xy$ (d) $6a$ and $6b$

Part B — Collecting, Adding and Subtracting

6. Simplify each expression by collecting like terms.
(a) $5x + 8 - 3x + 2$ (b) $9y^2 - 4y + 3y^2 + 6y - 5$

(c) $4a - 7b + 3a + 2b - 1$ (d) $8x^2 - 3x + 5 - 6x^2 - x + 9$

(e) $-6 + 2x^2 - 9x + 11 - 5x^2 + 4x$

7. Add each pair of polynomials (horizontal method).
(a) $(3x^2 + 2x - 5) + (x^2 - 6x + 4)$ (b) $(5a - 2b + 3) + (-a + 4b - 7)$

8. Add using the vertical method:

$$\begin{array}{r} 6x^2 - 5x + 2 \\ +\ (2x^2 + 3x - 8) \\ \hline \end{array}$$

9. Subtract each pair of polynomials (horizontal method).

(a) $(7x^2 + 4x - 3) - (2x^2 + 6x - 5)$

(b) $(5y^2 - 3y + 8) - (-2y^2 + 4y - 6)$

10. Subtract using the vertical method:

$$\begin{array}{r} 8x^2 - 2x + 6 \\ -\ (3x^2 + 5x - 4) \\ \hline \end{array}$$

Part C — Putting It Together

11. Simplify: $(4x^2 + 3x - 2) + (5x - 7) - (2x^2 - x + 6)$.

12. A student simplified $(6x^2 + 5x - 3) - (2x^2 - 4x + 1)$ as $6x^2 + 5x - 3 - 2x^2 - 4x + 1 = 4x^2 + x - 2$. Identify the student's mistake, and give the correct simplified answer.

13. A triangle has a perimeter of $(9x + 4)$ cm. Two of its sides measure $(3x - 2)$ cm and $(2x + 5)$ cm. Write and simplify an expression for the length of the third side.

14. *(Multiple Choice)* Which expression is the correct simplification of $(5x^2 - 3x + 1) - (2x^2 + 4x - 6)$? A. $3x^2 - 7x + 7$ B. $3x^2 + x - 5$ C. $3x^2 - 7x - 5$ D. $7x^2 + x + 7$

15. *(Numerical Response)* Simplify $(4x^2 + 9x - 5) - (x^2 + 3x - 2) + (x^2 - 6)$, and state the coefficient of the x term in the simplified answer.

Answer Key. 1. (a) binomial (b) monomial (c) trinomial. **2.** (a) 1 (b) 2 (c) 0 (d) 2 (e) 3. **3.** (a) 4 terms (b) 6, -4, 1 (c) -8 (d) degree 3. **4.** x: $3x$, x; x^2: $-5x^2$, $-2x^2$; y: $4y$, $6y$; constants: 8, -9. **5.** (a) like (b) not like — exponents differ (2 vs. 1) (c) like (d) not like — different variables. **6.** (a) $2x + 10$ (b) $12y^2 + 2y - 5$ (c) $7a - 5b - 1$ (d) $2x^2 - 4x + 14$ (e) $-3x^2 - 5x + 5$. **7.** (a) $4x^2 - 4x - 1$ (b) $4a + 2b - 4$. **8.** $8x^2 - 2x - 6$. **9.** (a) $5x^2 - 2x + 2$ (b) $7y^2 - 7y + 14$. **10.** $5x^2 - 7x + 10$. **11.** $2x^2 + 9x - 15$. **12.** Only the $2x^2$ sign was flipped; $-(2x^2 - 4x + 1) = -2x^2 + 4x - 1$, so the answer is $4x^2 + 9x - 4$. **13.** $(9x + 4) - (3x - 2) - (2x + 5) = 4x + 1$, so the third side is $(4x + 1)$ cm. **14.** A. **15.** $4x^2 + 6x - 9$; the coefficient of x is 6.

Lesson 2 Assignment *Show your work.*

Part A — Monomials and Binomials

1. Multiply the monomials.
(a) $3x^2 \times 4x^3$ (b) $-2x \times 6x^4$ (c) $5x^3 \times x^2$ (d) $-4x^2 \times (-3x^5)$

2. Use the distributive property to expand.
(a) $5(x+4)$ (b) $3(x-6)$ (c) $-2(x+9)$ (d) $-7(x-1)$

3. Expand.
(a) $3x(x+5)$ (b) $2x(x-8)$ (c) $-4x(x+2)$ (d) $-x(x-6)$

Part B — Trinomials and Collecting Like Terms

4. Expand.
(a) $4(x^2+3x+2)$ (b) $-3(x^2-2x+5)$ (c) $2x(x^2+4x-1)$ (d) $-5x(x^2-x+3)$

5. Expand and simplify.
(a) $3(x+2)+4x$ (b) $5x+2(x-3)$ (c) $6(x-1)-2x$ (d) $4-3(x+5)$

6. Expand and simplify.
(a) $2x(x+3)+5x$ (b) $3x(x-4)-2x$ (c) $x(x+7)-3(x+1)$ (d) $2(x-3)+x(x+5)$

7. Identify the error in each simplification below and give the correct simplification.
(a) $3x(2x+5) = 6x+15$ (b) $x^2(x^3-4x) = x^6-4x^3$

(c) $4(x-2)-3(x-2) = 4x-8-3x-6 = x-14$

Part C — Two Distributions and Applications

8. Expand and simplify.
(a) $-3x(x+4)-2(x-5)$ (b) $-2x(x-3)-5x(x+1)$ (c) $4x(2x-1)-3x(x-2)$ (d) $-x(x+6)+2x(x-3)$

9. A rectangle has width $3x$ and length $(x+8)$. Write a simplified expression for its area.

10. *(Multiple Choice)* Which expression is equivalent to $-4x(x-5)$? A. $-4x^2-20x$ B. $-4x^2+20x$ C. $4x^2-20x$ D. $4x^2+20x$

11. *(Multiple Choice)* Which expansion below is *incorrect*? A. $2x(3x+4) = 6x^2+8x$ B. $-5(x-3) = -5x+15$ C. $-3x(x+2) = -3x^2-6x$ D. $x(x-9) = x^2-81$

12. *(Numerical Response)* The expression $2x(x+3) - x(x-4)$ can be written in the form $ax^2 + bx$. Determine the value of $a + b$.

13. Expand and simplify.
(a) $5x(x+2) - 3x(x-1)$

(b) $-2x(3x-4) + 5(x+2)$

14. Expand and simplify: $-3x(2x+1) - 2x(x-5)$.

15. *(Numerical Response)* The expression $-4x(x+2) + 3x(x-5)$ can be written in the form $ax^2 + bx$. Determine the value of $a \times b$.

Answer Key. 1. (a) $12x^5$ (b) $-12x^5$ (c) $5x^5$ (d) $12x^7$. **2.** (a) $5x + 20$ (b) $3x - 18$ (c) $-2x - 18$ (d) $-7x + 7$. **3.** (a) $3x^2 + 15x$ (b) $2x^2 - 16x$ (c) $-4x^2 - 8x$ (d) $-x^2 + 6x$. **4.** (a) $4x^2 + 12x + 8$ (b) $-3x^2 + 6x - 15$ (c) $2x^3 + 8x^2 - 2x$ (d) $-5x^3 + 5x^2 - 15x$. **5.** (a) $7x + 6$ (b) $7x - 6$ (c) $4x - 6$ (d) $-3x - 11$. **6.** (a) $2x^2 + 11x$ (b) $3x^2 - 14x$ (c) $x^2 + 4x - 3$ (d) $x^2 + 7x - 6$. **7.** (a) the x of $3x$ was dropped from both products ($6x^2$, $15x$); $6x^2 + 15x$ (b) $x^2 \times x^3 = x^5$, not x^6; $x^5 - 4x^3$ (c) $-3(x-2) = -3x + 6$, not $-3x - 6$; $x - 2$. **8.** (a) $-3x^2 - 14x + 10$ (b) $-7x^2 + x$ (c) $5x^2 + 2x$ (d) $x^2 - 12x$. **9.** $3x^2 + 24x$. **10.** B. **11.** D. **12.** 11. **13.** (a) $2x^2 + 13x$ (b) $-6x^2 + 13x + 10$. **14.** $-8x^2 + 7x$. **15.** 23.

Lesson 3 Assignment *Show your work.*

Part A — Distributing and FOIL

1. *(Recall)* Use the distributive property to expand.
(a) $4(x+6)$ (b) $-3(x+5)$ (c) $2x(x+7)$ (d) $-x(x-8)$

2. Show *both* distributing steps — distribute the whole first binomial, then distribute each of its terms — before writing the final simplified answer.
(a) $(x+4)(x+5)$ (b) $(x+7)(x+2)$

3. Use the distributive property or FOIL to determine each product.
(a) $(x+3)(x+8)$ (b) $(x+9)(x+4)$ (c) $(x+2)(x+10)$

(d) $(x+6)(x+7)$

4. Use FOIL to determine each product.
(a) $(2x+3)(x+4)$ (b) $(3x+2)(x+5)$ (c) $(4x+1)(x+3)$

(d) $(5x+2)(x+1)$

5. Use FOIL to determine each product.
(a) $(2x+3)(3x+1)$ (b) $(3x+4)(2x+5)$ (c) $(4x+1)(2x+3)$

(d) $(5x+2)(2x+1)$

Part B — Negative Terms and Two Variables

6. Expand each product. Watch your signs carefully.
(a) $(x-3)(x+7)$ (b) $(x+5)(x-4)$ (c) $(x-2)(x-9)$

(d) $(2x-1)(x+6)$ (e) $(3x+2)(x-5)$ (f) $(2x-3)(3x-2)$

7. Expand each product, involving two variables.
(a) $(x+2y)(x+5y)$ (b) $(x+4y)(x-3y)$ (c) $(2x+y)(x+3y)$

8. Expand and simplify.

(a) $(3x-4)(2x+5)$ (b) $(4x+3)(3x-2)$ (c) $(5x-2)(2x-3)$ (d) $(3x-1)(4x-5)$

9. Identify the error in each expansion and give the correct result.
(a) $(x+5)(x+3) = x^2+15$ (b) $(x-2)(x+7) = x^2+9x-14$ (c) $(2x-3)(x-4) = 2x^2-11x-12$

Part C — Expanding and Simplifying Further

10. Expand and simplify.
(a) $(x+3)(x+4)+(x+2)(x+5)$ (b) $(x+6)(x+2)+(x+1)(x+3)$

(c) $(x-2)(x+5)+(x+3)(x-1)$

11. Expand and simplify.
(a) $(x+7)(x+4)-(x+3)(x+2)$ (b) $(x+5)(x+6)-(x+2)(x+4)$

(c) $(x+8)(x-1)-(x-2)(x+3)$

12. *(Multiple Choice)* Which of the following is the correct expansion of $(2x-3)(x+4)$? A. $2x^2+5x-12$ B. $2x^2+11x-12$ C. $2x^2-5x-12$ D. $2x^2+5x+12$

13. *(Multiple Choice)* Which of the following expansions is *incorrect*?
A. $(x+4)(x+3) = x^2+7x+12$ B. $(x-5)(x+2) = x^2-3x-10$
C. $(2x+1)(x-3) = 2x^2-5x-3$ D. $(3x-2)(x+5) = 3x^2+13x+10$

14. *(Numerical Response)* The expression $(x+5)(x+2)+(x+3)(x+1)$ can be written in the form ax^2+bx+c. Determine the value of $a+b+c$.

15. A rectangular deck has length $(x+8)$ m and width $(x+5)$ m. Determine an expression for its area, in m^2, by (i) using the distributive property, and (ii) using FOIL. Confirm that both methods agree.

16. *(Numerical Response)* The area of a rectangle with length $(x+1)$ cm and width $(x+6)$ cm can be written as x^2+px+q, where p and q are natural numbers less than 10. Record the value of p in the first box and q in the second box, left to right.

Answer Key. 1. (a) $4x+24$ (b) $-3x-15$ (c) $2x^2+14x$ (d) $-x^2+8x$. **2.** (a) $x^2+9x+20$ (b) $x^2+9x+14$. **3.** (a) $x^2+11x+24$ (b) $x^2+13x+36$ (c) $x^2+12x+20$ (d) $x^2+13x+42$. **4.** (a) $2x^2+11x+12$ (b) $3x^2+17x+10$ (c) $4x^2+13x+3$ (d) $5x^2+7x+2$. **5.** (a) $6x^2+11x+3$ (b) $6x^2+23x+20$ (c) $8x^2+14x+3$ (d) $10x^2+9x+2$. **6.** (a) $x^2+4x-21$ (b) x^2+x-20 (c) $x^2-11x+18$ (d) $2x^2+11x-6$ (e) $3x^2-13x-10$ (f) $6x^2-13x+6$. **7.** (a) $x^2+7xy+10y^2$ (b) $x^2+xy-12y^2$ (c) $2x^2+7xy+3y^2$. **8.** (a) $6x^2+7x-20$ (b) $12x^2+x-6$ (c) $10x^2-19x+6$ (d) $12x^2-19x+5$. **9.** (a) O and I terms missed; $x^2+8x+15$ (b) middle terms combined with wrong signs ($7x-2x$); $x^2+5x-14$ (c) sign error on Last ($-3\times-4=+12$); $2x^2-11x+12$. **10.** (a) $2x^2+14x+22$ (b) $2x^2+12x+15$ (c) $2x^2+5x-13$. **11.** (a) $6x+22$ (b) $5x+22$ (c) $6x-2$. **12.** A. **13.** D. **14.** 26. **15.** $(x+8)(x+5)=x^2+13x+40$ m^2; both methods agree. **16.** 7 6 ($p=7$, $q=6$).

Lesson 4 Assignment *Show your work.*

Part A — Expand and Simplify

Use $(a+b)(a-b) = a^2 - b^2$ or $(a \pm b)^2 = a^2 \pm 2ab + b^2$ wherever the pattern applies.

1. $(x+3)(x-3)$
2. $(x-7)(x+7)$
3. $(x+4)^2$
4. $(x-5)^2$
5. $(x+9)(x-9)$
6. $(x+6)^2$
7. $(x-6)^2$
8. $(2x+3)(2x-3)$
9. $(3x+1)^2$
10. $(5x-1)^2$
11. $(6-x)(6+x)$
12. $(2x+7)^2$
13. $(5x-2)^2$
14. $(x+6)(x-2)$ *(Careful — does either special pattern apply here?)*

Part B — Reasoning

15. *(Multiple Choice)* A student expanded $(x+8)^2$ and wrote $(x+8)^2 = x^2 + 64$. Which statement correctly describes the student's mistake? A. The student forgot the middle term $2ab = 16x$; the correct expansion is $x^2 + 16x + 64$. B. The student computed 8^2 incorrectly. C. The student used the difference of squares pattern instead of the perfect square pattern. D. The student forgot to square the first term x.

16. *(Numerical Response)* Given that $(x+a)(x-a) = x^2 - 64$ for every value of x and that $a > 0$, the value of a is _____.

Answer Key. 1. $x^2 - 9$ **2.** $x^2 - 49$ **3.** $x^2 + 8x + 16$ **4.** $x^2 - 10x + 25$ **5.** $x^2 - 81$ **6.** $x^2 + 12x + 36$ **7.** $x^2 - 12x + 36$ **8.** $4x^2 - 9$ **9.** $9x^2 + 6x + 1$ **10.** $25x^2 - 10x + 1$ **11.** $36 - x^2$ **12.** $4x^2 + 28x + 49$ **13.** $25x^2 - 20x + 4$ **14.** $x^2 + 4x - 12$ (neither pattern — FOIL) **15.** A (correct expansion is $x^2 + 16x + 64$) **16.** $a = 8$.

Lesson 5 Assignment *Show your work.*

Part A — Six Partial Products, then Collect

Expand and simplify questions 1–6. Use a grid or the distributive property.

1. $(x+1)\left(x^2+2x+3\right)$

2. $(x+4)\left(x^2+x+2\right)$

3. $(2x+1)\left(x^2+3x+2\right)$

4. $(3x+2)\left(x^2+x+1\right)$

5. $(x+5)\left(x^2+2x+6\right)$

6. $(2x+3)\left(x^2+4x+1\right)$

Part B — Products with Negative Terms

Expand and simplify questions 7–12. Check the sign of every partial product.

7. $(x-2)\left(x^2+3x+4\right)$

8. $(x-5)\left(x^2-2x+3\right)$

9. $(2x-1)\left(x^2+2x-3\right)$

10. $(3x-2)\left(x^2-x+4\right)$

11. $(x-4)\left(x^2-3x-2\right)$

12. $(2x-5)\left(x^2-x-2\right)$

Part C — Combining and Extending

Expand and simplify questions 13–14.

13. $(x+2)\left(x^2+3x+1\right)+\left(x^2-4x+5\right)$

14. $(x-3)\left(x^2+x+2\right)-\left(2x^2-3x+1\right)$

15. *(Multiple Choice)* A student expands $(x+3)\left(x^2+2x+1\right)$ as shown. Which statement is true?

$(x+3)\left(x^2+2x+1\right)$
$= x\left(x^2+2x+1\right)+3\left(x^2+2x+1\right)$ Line 1
$= x^3+2x^2+x+3x^2+6x+3$ Line 2
$= x^3+5x^2+6x+3$ Line 3

A. Error in Line 1. B. Error in Line 2.
C. Error in Line 3. D. The expansion is correct.

16. *(Numerical Response)* Adding the product of $(x+3)$ and $\left(x^2+2x-1\right)$ to the product of $(x-1)$ and $\left(x^2-x+4\right)$ gives a polynomial of the form ax^3+bx^2+cx+d. The value of $a+b$ is _____.

Answer Key. 1. x^3+3x^2+5x+3 **2.** x^3+5x^2+6x+8 **3.** $2x^3+7x^2+7x+2$ **4.** $3x^3+5x^2+5x+2$ **5.** $x^3+7x^2+16x+30$ **6.** $2x^3+11x^2+14x+3$ **7.** x^3+x^2-2x-8 **8.** $x^3-7x^2+13x-15$ **9.** $2x^3+3x^2-8x+3$ **10.** $3x^3-5x^2+14x-8$ **11.** $x^3-7x^2+10x+8$ **12.** $2x^3-7x^2+x+10$ **13.** x^3+6x^2+3x+7 **14.** x^3-4x^2+2x-7 **15.** C (middle terms give $x+6x=7x$, so Line 3 should be x^3+5x^2+7x+3) **16.** 5.

Lesson 6 Assignment *Show your work.*

Part A — Area and Volume of Simple Shapes

1. A rectangular window has length $(x + 3)$ ft and width $(x + 1)$ ft. (a) Write the area as a simplified trinomial. (b) Determine the area when $x = 5$.

2. A rectangular patio has length $(x + 9)$ m and width $(x + 2)$ m. (a) Write the area as a simplified trinomial. (b) Determine the area when $x = 3$.

3. A rectangular shipping box has length $(x+1)$ cm, width $(x+3)$ cm, and height $(x+5)$ cm. (a) Write the volume as a simplified polynomial. (b) Determine the volume when $x = 3$.

4. A rectangular crate has a square base with side length $(x + 2)$ m and a height of 4 m. (a) Write and simplify an expression for the volume. (b) Determine the volume when $x = 3$.

Part B — Composite Areas and Equal Areas

5. A rectangular picture, including its wooden frame, has outer dimensions $(x + 8)$ cm by $(x + 6)$ cm. The opening inside the frame is a square of side length $(x + 2)$ cm. (a) Write a simplified expression for the area of the wooden frame itself. (b) Evaluate it when $x = 3$.

6. A rectangular lawn measures $(x + 10)$ m by $(x + 6)$ m. A square flower bed of side length $(x + 3)$ m is cut out of one corner. (a) Write a simplified expression for the area of grass remaining. (b) Evaluate the area of grass when $x = 4$.

7. Rectangle C measures $(x + 3)$ by $(x + 7)$. Rectangle D measures $(x + 2)$ by $(x + 9)$. The two rectangles have exactly the same area. (a) Write an equation setting the area of C equal to the area of D. (b) Expand both sides and simplify to solve for x. (c) State the area of each rectangle.

8. A square has side length $(x + 4)$ m. A rectangle has dimensions $(x + 2)$ m by $(x + 7)$ m. The square and the rectangle have the same area. Determine the value of x.

Part C — Multi-Step Applications

9. A rectangular planter box has a base with length $(x+3)$ ft and width $(x+2)$ ft, and a height of $(x+4)$ ft. (a) Write and simplify an expression for the area of the base. (b) Write and simplify an expression for the volume. (c) If $x = 2$, calculate the volume.

10. A rectangular sandbox has a base with length $(x + 5)$ ft and width $(x+2)$ ft, and a depth of $(x+1)$ ft. A wooden cover is built 2 ft longer and 2 ft wider than the base. (a) Area of the base. (b) Volume of sand needed to fill it. (c) Area of the cover. (d) If $x = 2$, calculate the volume of sand and the area of the cover.

11. *(Multiple Choice)* Which expression represents the area of a rectangle with side lengths $(x+6)$ and $(x-2)$?
A. $x^2+4x-12$ B. $x^2+8x-12$ C. $x^2-4x-12$ D. $x^2+4x+12$

12. *(Numerical Response)* A rectangular garden bed has length $(x+11)$ m and width $(x+4)$ m. If $x=6$, the area of the garden bed, in square metres, is _____.

13. A rectangular vegetable garden measures $(x+8)$ m by $(x+3)$ m. A gravel path of uniform width surrounds the garden, so the total outer dimensions (garden plus path) are $(x+12)$ m by $(x+7)$ m. (a) Determine a simplified expression, in terms of x, for the area of the path alone. (b) Evaluate the area of the path when $x=2$.

Answer Key. 1. (a) x^2+4x+3 ft^2 (b) 48 ft^2. **2.** (a) $x^2+11x+18$ m^2 (b) 60 m^2. **3.** (a) $x^3+9x^2+23x+15$ cm^3 (b) 192 cm^3. **4.** (a) $4x^2+16x+16$ m^3 (b) 100 m^3. **5.** (a) $10x+44$ cm^2 (b) 74 cm^2. **6.** (a) $10x+51$ m^2 (b) 91 m^2. **7.** (a) $(x+3)(x+7)=(x+2)(x+9)$ (b) $x=3$ (c) each has area 60 square units. **8.** $x=2$ (each area is 36 m^2). **9.** (a) x^2+5x+6 ft^2 (b) $x^3+9x^2+26x+24$ ft^3 (c) 120 ft^3. **10.** (a) $x^2+7x+10$ ft^2 (b) $x^3+8x^2+17x+10$ ft^3 (c) $x^2+11x+28$ ft^2 (d) volume $=84$ ft^3, cover area $=54$ ft^2. **11.** A. **12.** 170. **13.** (a) $8x+60$ m^2 (b) 76 m^2.

Lesson 7 Assignment *Show your work.*

Part A — Like Terms, Adding and Subtracting

1. Simplify by collecting like terms: $4x^2 + 9x - 6 - 2x^2 - 3x + 10$

2. Simplify: $7m - 4n - 3m + 9n - 2$

3. Add: $(5x^2 - 2x + 6) + (3x^2 + 7x - 9)$

4. Subtract: $(8x^2 + 3x - 5) - (2x^2 - 6x + 9)$

Part B — Multiplying a Monomial by a Polynomial

5. Multiply: $-5x^3 \times 4x^2$

6. Expand: $3x(2x - 7)$

7. Expand: $-4x\left(x^2 + 3x - 2\right)$

8. Expand and simplify: $5x(x + 4) - 3x(x - 2)$

Part C — Multiplying Two Binomials

9. Expand using FOIL: $(x + 6)(x - 4)$

10. Expand: $(2x - 5)(3x + 2)$

11. Expand and simplify: $(x + 3)(x + 8) + (x - 2)(x + 5)$

Part D — Special Products

12. Expand using the difference of squares pattern: $(6x + 1)(6x - 1)$

13. Expand using the perfect square trinomial pattern: $(x + 9)^2$

14. Expand: $(3x - 5)^2$

15. Identify which pattern applies, then expand: $(7 - 2x)(7 + 2x)$

Part E — Multiplying a Binomial by a Trinomial

16. Expand: $(x + 4)\left(x^2 - 3x + 5\right)$

17. Expand: $(2x - 1)\left(x^2 + 4x - 3\right)$

Part F — Area and Volume Applications

18. A rectangular playing field has length $(x + 9)$ m and width $(x + 5)$ m. Write a simplified expression for its area.

19. A storage bin shaped like a rectangular prism has a square base of side length $(x + 3)$ cm and height x cm. Write a simplified expression for its volume.

20. A square poster of side length $(x+6)$ cm has a smaller square of side length x cm cut from one corner. Write a simplified expression for the remaining area.

Part G — Test-Style Questions

21. *(Numerical Response)* The expression $3x(x+5) - 2\left(x^2 - 4x + 1\right)$ can be written in the form $ax^2 + bx + c$. The value of $a + b + c$ is _____.

22. *(Written Response)* Malik and Sana each attempt to expand and simplify $4(x-2)^2 - 3x$. Their work is shown below, and each student's work contains exactly one mathematical error.

Malik	Sana
$4(x-2)^2 - 3x$	$4(x-2)^2 - 3x$
$(4x-8)^2 - 3x$	$4(x^2 - 2x + 4) - 3x$
$16x^2 - 64x + 64 - 3x$	$4x^2 - 8x + 16 - 3x$
$16x^2 - 67x + 64$	$4x^2 - 11x + 16$

(a) Describe in detail the error Malik made. (b) Describe in detail the error Sana made. (c) Determine the correct expansion of $4(x-2)^2 - 3x$. (d) Verify your expansion by substituting $x = 5$ into both the original expression and your expansion, confirming the same value.

Answer Key. **1.** $2x^2 + 6x + 4$. **2.** $4m + 5n - 2$. **3.** $8x^2 + 5x - 3$. **4.** $6x^2 + 9x - 14$. **5.** $-20x^5$. **6.** $6x^2 - 21x$. **7.** $-4x^3 - 12x^2 + 8x$. **8.** $2x^2 + 26x$. **9.** $x^2 + 2x - 24$. **10.** $6x^2 - 11x - 10$. **11.** $2x^2 + 14x + 14$. **12.** $36x^2 - 1$. **13.** $x^2 + 18x + 81$. **14.** $9x^2 - 30x + 25$. **15.** Difference of squares; $49 - 4x^2$. **16.** $x^3 + x^2 - 7x + 20$. **17.** $2x^3 + 7x^2 - 10x + 3$. **18.** $x^2 + 14x + 45$ m^2. **19.** $x^3 + 6x^2 + 9x$ cm^3. **20.** $12x + 36$ cm^2. **21.** 22. **22.** Malik squared $4(x-2)$ instead of multiplying by 4 after squaring; Sana expanded $(x-2)^2$ as $x^2 - 2x + 4$ (middle term should be $-4x$, and $4(x-2)^2 = 4x^2 - 16x + 16$). Correct: $4x^2 - 19x + 16$; both give 21 at $x = 5$.